

Week Nine Technical Report

Redux Toolkit, Authentication, and Dashboard Development

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Teach2Give QA & QE Training Program

Overview

Week 9 focused on advanced state management and authentication using Redux Toolkit (RTK) and Redux Toolkit Query (RTK Query). The main topics included user authentication, role-based access control, dashboard development for admins and users, and Todo management. These sessions built upon previous React knowledge and marked a significant step towards creating fully functional and interactive applications with persistent state and secure access controls.

Training Modules Covered

We began by introducing Redux Toolkit and RTK Query to manage state and connect the frontend to backend APIs efficiently. Key learnings included creating Redux slices, handling API requests for user registration and login, and persisting authenticated user data in local storage.

Next, we implemented the authentication workflow which include user registration, user verification and user login.

We also covered role-based access management which included assigning roles and implementing role based redirection and access control ensuring admins and users only access their respective dashboards.

Finally, we integrated Todo management which included CRUD operations for the Todo items, ensuring users and admins have their respective access on what they can implement on the Todos, combining all Todo functionalities in their respective dashboards.

Practical Exercise and Project

The practical exercises involved building and testing a fully functional Admin-User Dashboard. This included ensuring smooth login and registration flows using RTK Query, verifying that users could be authenticated and their data persisted in Redux and local storage, and testing role-based access control by logging in as admins and users to confirm correct redirection and access restrictions. Additionally, complete CRUD functionality for Todos was implemented, connected to user sessions, and the exercises laid the foundation for future enhancements in dashboard management.

Challenges Faced

The main challenges I faced this week were dealing with bugs in the code, connecting the frontend and backend seamlessly, and catching up with the pace of learning Redux Toolkit and integrating it into the application. These difficulties were resolved through careful debugging, testing API connections, and gradually understanding how to manage state and persist user data effectively across components.

Recommendations

To overcome these challenges, I recommend debugging and testing the connection between the frontend and backend, ensuring API requests and responses are correctly handled. Consulting with peers or the trainer when encountering persistent bugs can save time, as others may have faced similar issues and can provide guidance or solutions. Additionally, practicing state management and persistence in Redux through small exercises will build confidence and reduce errors in future development.

Conclusion

This week solidified my understanding of advanced state management and authentication in React applications using Redux Toolkit. By building Admin-User dashboards with persistent authentication, role-based access control, and integrated Todo management, I gained practical experience in developing scalable and secure applications, forming a strong foundation for upcoming project work.

Signature:

David Karanja